ELEMENTS OF A ONE-DIMENSIONAL NOETHERIAN LOCAL RING THAT AGREE IN EVERY MINIMAL-PRIME LOCALIZATION AGREE IN SOME POWER

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof cites no external \ref inside the counted span, so the assembly needs no external-reference stubs.

Allowed standard prerequisites (editorial).

- Nilradical as intersection of minimal primes; annihilator and support. Stacks Project, Algebra (prime ideals, radicals, and annihilators in Noetherian rings). Supplies the facts that intersection of minimal primes is the nilradical and that a prime avoids the annihilator I exactly when it lies over it.

Lemma 0.1. *Let A be a Noetherian local ring of dimension 1. Let $L = \prod A_{\mathfrak{p}}$ where the product is over the minimal primes of A. Let $a_1, a_2 \in \mathfrak{m}_A$ map to the same element of L. Then $a_1^n = a_2^n$ for some $n > 0$.*

Proof. Write $a_1 = a_2 + x$. Then x maps to zero in L. Hence x is a nilpotent element of A because $\bigcap \mathfrak{p}$ is the radical of (0) and the annihilator I of x contains a power of the maximal ideal because $\mathfrak{p} \notin V(I)$ for all minimal primes. Say $x^k = 0$ and $\mathfrak{m}^n \subset I$. Then

$$a_1^{k+n} = a_2^{k+n} + \binom{n+k}{1} a_2^{n+k-1}x + \binom{n+k}{2} a_2^{n+k-2}x^2 + \ldots + \binom{n+k}{k-1} a_2^{n+1}x^{k-1} = a_2^{n+k}$$

because $a_2 \in \mathfrak{m}_A$. □

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